PARTH JAIN

IIT Madras B.S. Degree

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SUMMARY

Computer Science student building full-stack systems, AI/ML pipelines, and production-grade applications. Strong academic record across VIT Bhopal and IIT Madras BS Data Science, with experience in spanning full-stack web development, backend systems engineering, and modern ML tooling.

EDUCATION

Indian Institute of Technology — Chennai, Tamil Nadu
BS - Bachelors in Data Science and Applications — *May 2023 – May 2027*

Vellore Institute of Technology — Bhopal, Madhya Pradesh
B.Tech, Computer Science & Engineering — *Aug. 2023 – June 2027*

PROJECTS

Placement Portal System | *Flask, Vue.js, Redis, Celery, JWT, SQLAlchemy*

- Engineered a scalable multi-role placement platform with 40+ REST APIs covering student, recruiter, and admin workflows, reducing administrative manual effort by an estimated 30%.
- Architected a JWT-based Role-Based Access Control (RBAC) system ensuring 100% data isolation between student and recruiter accounts with secure session management and token refresh.
- Integrated Celery + Redis for asynchronous task processing, achieving 40% reduction in API response latency and 25% boost in responsiveness under concurrent load via strategic Redis caching.
- Implemented resume parsing (PyPDF2), PDF export (xhtml2pdf), automated email notifications (Flask-Mail), and AI-powered resume scoring via Gradio Client.

ML Classification Pipeline | *Python, scikit-learn, NLP, TF-IDF, Bayesian Tuning*

- Engineered an end-to-end ML pipeline to classify complex text patterns across a 200,000+ record dataset, leveraging 60,000-dimensional TF-IDF features for rich representation.
- Surpassed baseline to achieve an 0.826 Kaggle-style benchmark score using a voting ensemble of 15+ diverse ML models (Logistic Regression, Random Forest, Gradient Boosting, SVM).
- Reduced overfitting by 12% through rigorous k-fold cross-validation combined with Bayesian hyperparameter optimization.

InventoryPro — Inventory Management System | *Next.js 16, React 19, TypeScript, Prisma ORM, Recharts*

- Developed a modern inventory management SaaS application with real-time tracking, an analytics dashboard, and rich data visualizations using Next.js App Router and TypeScript.
- Built type-safe data models with Prisma ORM, implemented server-side rendering and API routes with Next.js, and used SWR for client-side data fetching with automatic cache revalidation.
- Implemented JWT-based authentication using JOSE, custom hooks for state management, and Recharts for interactive stock trend visualization.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS

Frameworks: Node.js, Flask, Bootstrap, FastAPI , Next.js, RESTful API design, JWT (auth), RBAC, Celery ,Redis

Developer Tools: Git, Docker, Vercel, VS Code, Linux, Google Colab, Render, Supabase, Hugging Face Spaces

Libraries: pandas, NumPy, Matplotlib, scikit-learn, TensorFlow/Keras, Transformers, RAG, Vector DBs

ML Techniques: Linear regression, Logistic Regression, Random Forest, SVM, Gradient Boosting, Voting Ensembles , k-fold CV , Bayesian Hyperparameter Tuning , TF-IDF, Prompt Engineering

CERTIFICATIONS

- Machine Learning Specialization | Coursera — DeepLearning.AI (Andrew Ng)
- Machine Learning | NPTEL, IIT Kharagpur
- Marketing Analytics | NPTEL, IIT Kharagpur
- Natural Language Processing with Transformers using Hugging Face | IIT Madras
- Introduction to HuggingFace and HuggingFace Spaces | IIT Madras
- The Bits and Bytes of Computer Networking | Google Career Certificates

EXTRA-CURRICULAR ACTIVITIES

- 800+ Day Coding Streak*— GeeksForGeeks; maintained consistent daily problem-solving discipline across 800+ consecutive days, demonstrating exceptional commitment to DSA mastery.
- Competitive Programming — Solved 100+ DSA problems on LeetCode and GeeksForGeeks, focusing on time & space complexity optimization across arrays, trees, graphs, dynamic programming, and backtracking.
- 2nd Rank Medal — State Level NCSC (National Children Science Congress); competed at state level in India's premier school-level science event.
- HuggingFace Deployment — Deployed Resume Analyzer & Matcher (Sentence Transformers) and NER Tool (BERT) on Hugging Face Spaces.
- Vercel Deployment – Deployed self made projects on vercel and personal portfolio
- Event Volunteer — AdVITya'24 (VIT Bhopal's annual technical fest); supported end-to-end event organization.